



Kingdom of Saudi Arabia
Ministry of Transport
General Directorate for Materials & Research
Quality Control Department

Fourier Transform Infrared Spectroscopy
(FT-IR)
For Handing Over Samples

FEBRUARY 2015

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Chapter 1

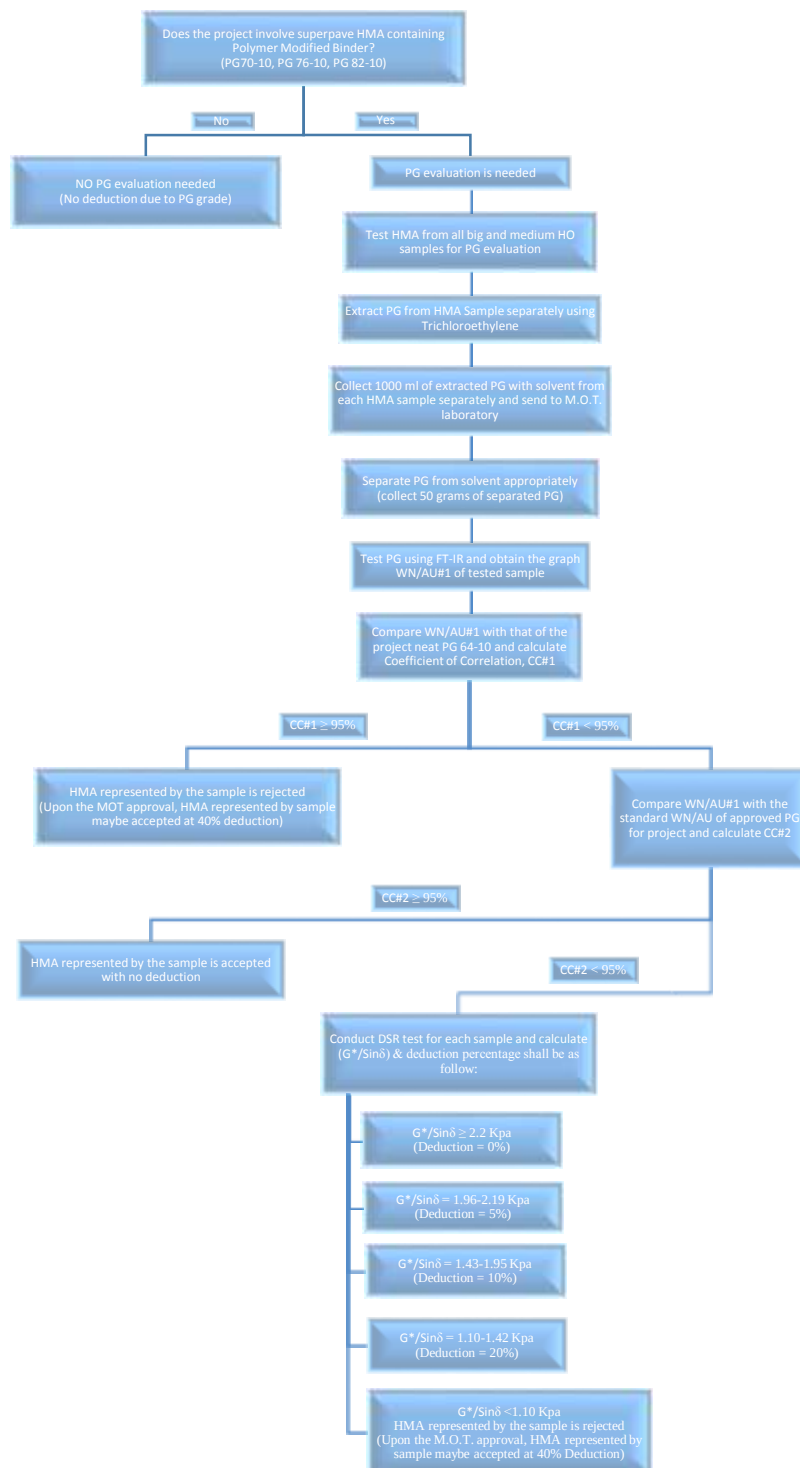
INTRODUCTION

Bruker Tensor 27 FT-IR Using PIKE MIRacle™ ATR cell

The Fourier Transform Infrared Spectroscopy (FT-IR) has the capability to measure the infrared light absorbed by a certain material. This absorption will depend on the chemical function present in the material and will result to a spectrum of absorbance vs wavenumber data given by the FT-IR program. ^[2]

The standard sample cell in the FT-IR is a PIKE MIRacle™ single-bounce attenuated total reflectance (ATR) cell equipped with a ZnSe single crystal. The sample that is placed on top of the ATR crystal can be solid, gel or non-corrosive solution. For solution samples like Trichloroethylene + Modified Bitumen, which is extracted from the handing over samples, the ATR cell is not very sensitive. Fairly concentrated samples are needed to get a semi-reasonable spectrum or results according to the Bruker Tensor 27 FT-IR Instruction Manual ^[1].

Deduction due to Deficiencies of Used PG Grade (Project Initial Handing Over)



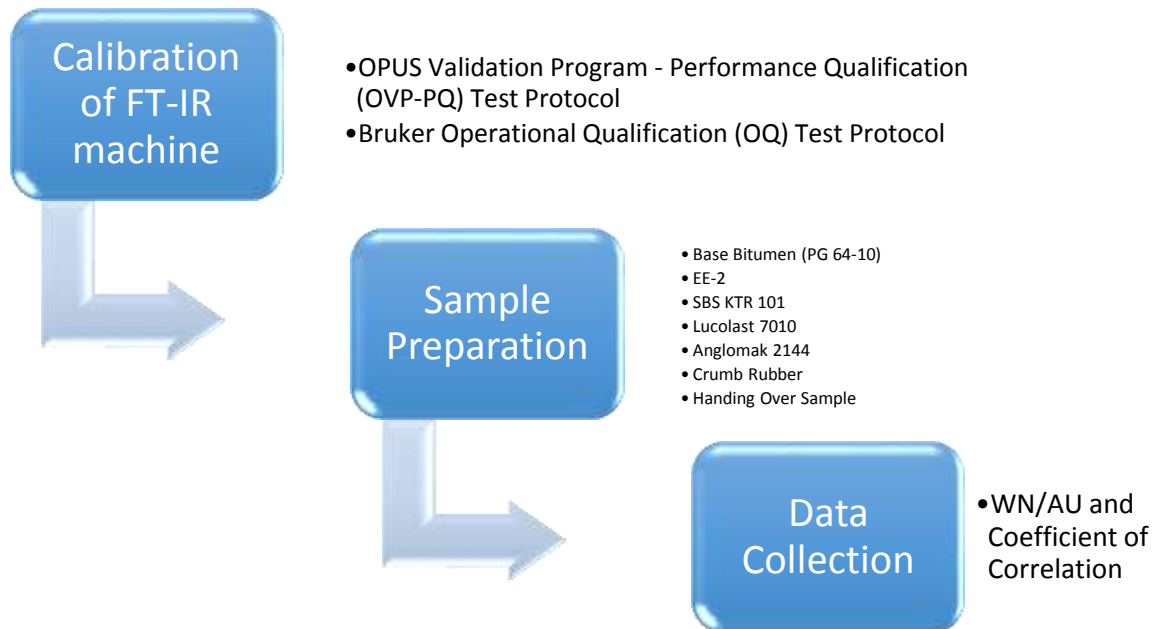
Procedures Conducted at Project Site

Procedures Conducted at the M.O.T. Central Laboratory

PROBLEM STATEMENT

1. Can the Bruker Tensor 27 FT-IR using PIKE MIRacle™ ATR cell detect the presence of the commonly used polymer types in PMABs?
2. Can the Bruker Tensor 27 FT-IR Using PIKE MIRacle™ ATR cell distinguish between the commonly used PMABs?
3. Can the Bruker Tensor 27 FT-IR Using PIKE MIRacle™ ATR cell determine the amount of polymer present in the PMABS?
3. Can the Bruker Tensor 27 FT-IR Using PIKE MIRacle™ ATR cell be used as a quality assurance test to detect the presence of polymer type and content in PMABs obtained from the plant tanks and from the Hot-Asphalt mix?

Chapter 2

METHODOLOGY**CALIBRATION OF THE FT-IR MACHINE**

The Bruker Tensor 27 FT-IR using OPUS 6.5 Data Collection program was calibrated by Mr. Kifayatullah Khalil, Sales & Service Engineer of National Scientific Company Limited, on February 19, 2015. Therefore, all official testing was done after that date.

The OPUS Validation Program (OVP) for the FT-IR allows to retrieve the parameter settings of the instrument. Measurement and test parameters are stored in a set of databases which have been installed when installing OPUS. This test must be run every three months.

The OQ (Operational Qualification) test protocol checks the instrument performance and compares this performance to the instrument specifications. OQ tests should normally run after installation, maintenance or major repair, exchange of optical components which may influence the instrument performance, on a regular yearly or semi-yearly basis. ^[3]

POLYMER INDICATORS

(Sample Preparation and FT-IR Settings)

To determine the presence of polymer with the Bruker Tensor 27 FT-IR using the PIKE MIRacle™ ATR cell, different 100 grams bitumen modified with 5.0%, 10.0% and 20.0% concentrations with the same base bitumen were prepared. Considering the filter paper during the PG extraction process of the Handing Over samples, only smaller particle sized Crumb Rubber were used (Passing # 50).

1. Base Bitumen (PG 64-10) from Ras Tanura Refinery
2. PG 64-10 from Ras Tanura Refinery Modified with EE-2
3. PG 64-10 from Ras Tanura Refinery Modified with SBS KTR 101
4. PG 64-10 from Ras Tanura Refinery Modified with Lucolast 7010
5. PG 64-10 from Ras Tanura Refinery Modified with Anglomaks 2144
6. PG 64-10 from Ras Tanura Refinery Modified with Crumb Rubber

Spectrum Acquisition Settings ^[2]

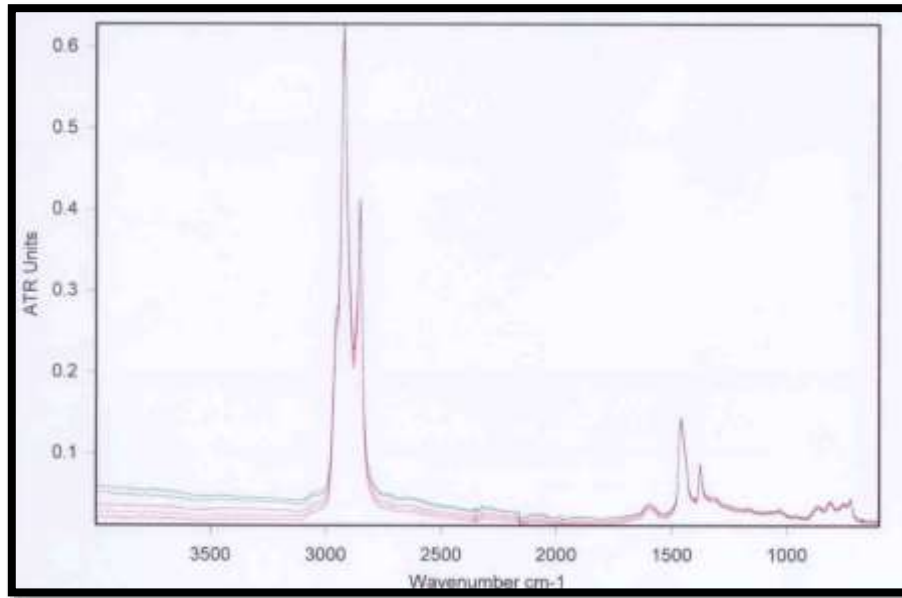
An average of 5 spectrum readings (WN/AU) are taken in order to produce a reasonable data. The spectral range is 4000 to 400 cm^{-1} with a resolution of 4 cm^{-1} . The acquisition time is 16 scans.

Standardization of FT-IR Spectrum ^[2]

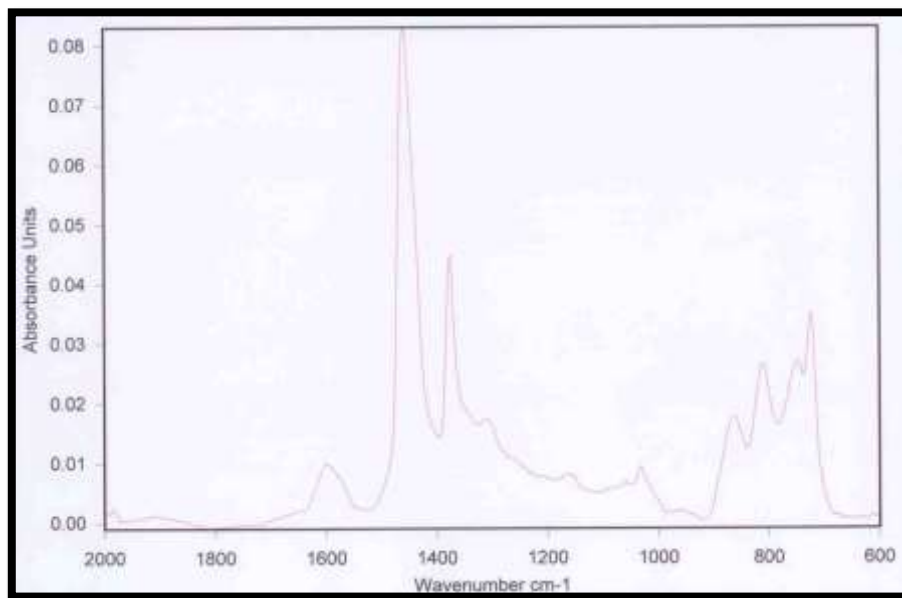
Recorded FT-IR spectrum are manipulated to allow practical comparison in the interesting part of the spectrum, as follows:

- Step 1 : Convert each ATR Spectrum to Absorbance Spectrum
- Step 2 : Cut each Spectrum range (2000 to 600 cm^{-1})
- Step 3 : Use Baseline Correction (Straight Line Option) where two lowest absorbance in the spectrum are joined to remove slopes in the zone of interest.
- Step 4 : Average the Absorbance Spectrum.

STANDARDIZATION OF RAS TANURA BASE BITUMEN

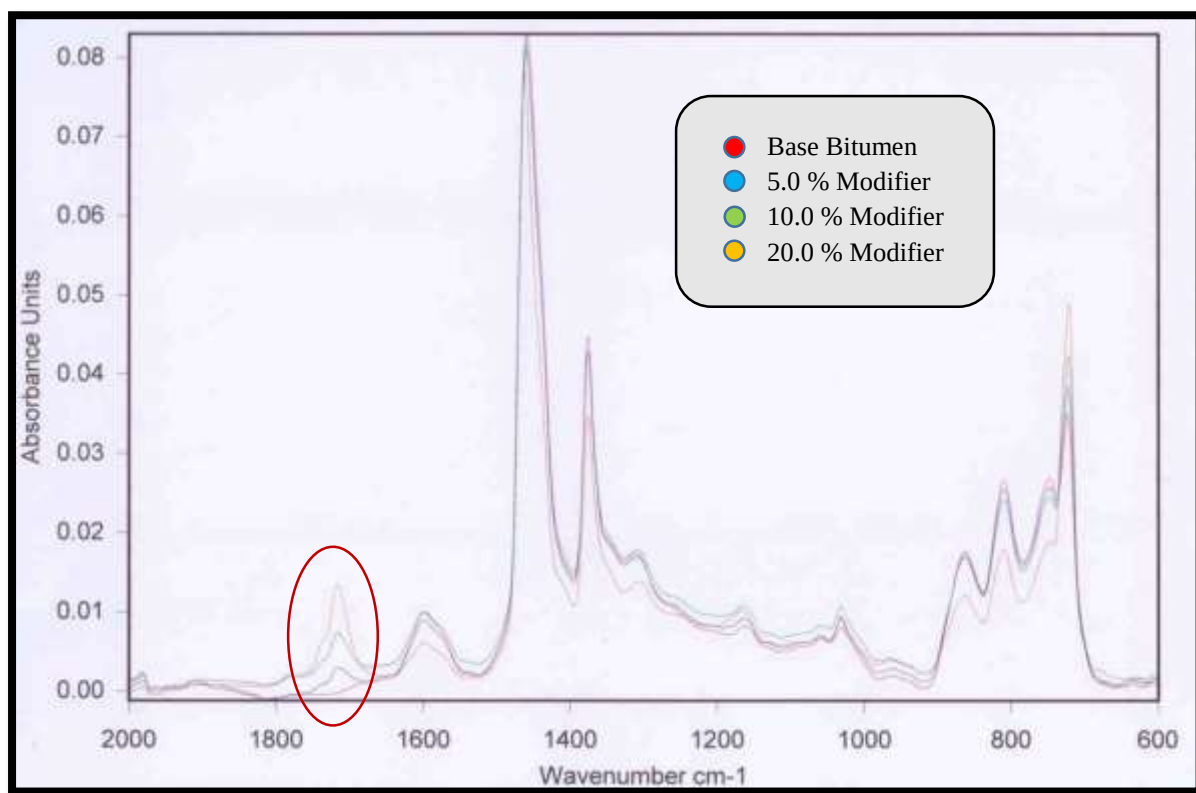


This WN/AU shows 5 ATR spectrum for the Base Bitumen of Ras Tanura.



This shows the average of the 5 standardized WN/AU Absorbance Spectrum.

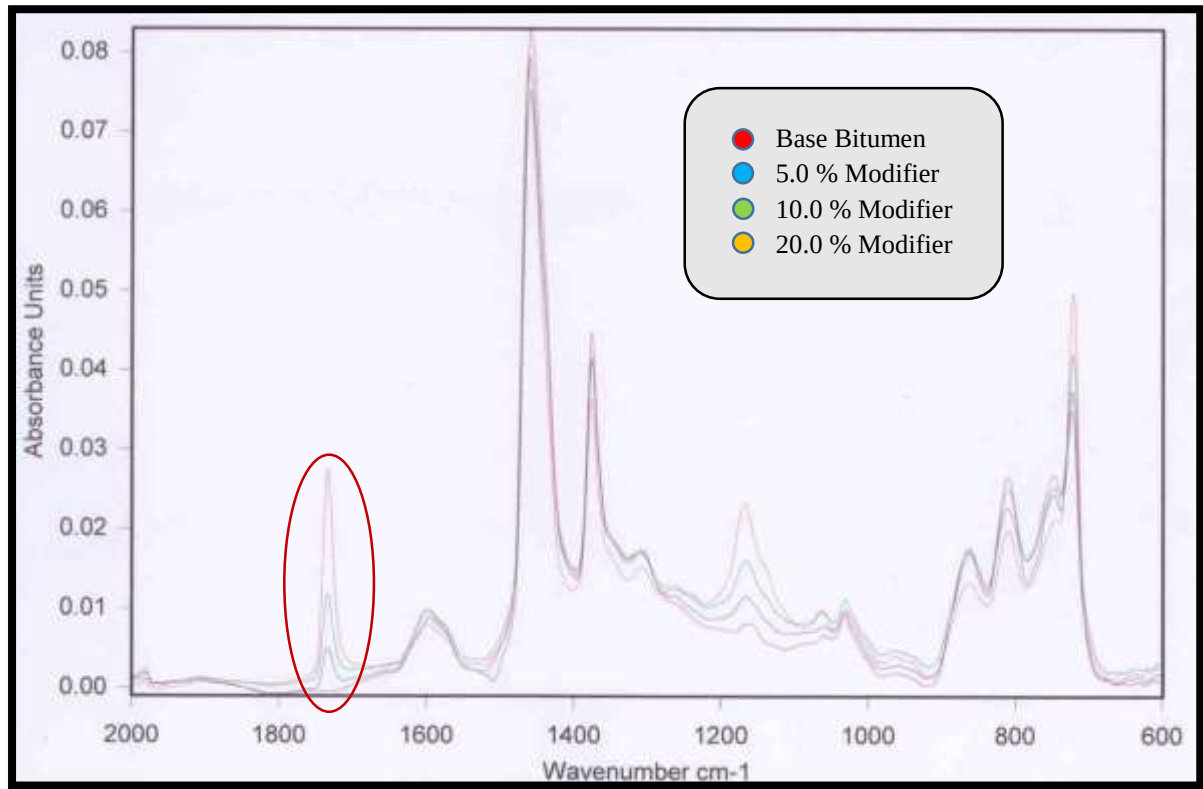
POLYMER INDICATOR FOR EE-2



For the Base Bitumen Modified with EE-2, the results show that the modifier affects the base bitumen at wavenumber 1717 cm^{-1} . As the amount of bitumen increases, the Absorbance Units also increases. The table shows the relationship between the Absorbance Units and the amount of Modifier.

Amount of Modifier, %	Absorbance Unit(AU)
5.0	0.003
10.0	0.007
20.0	0.014

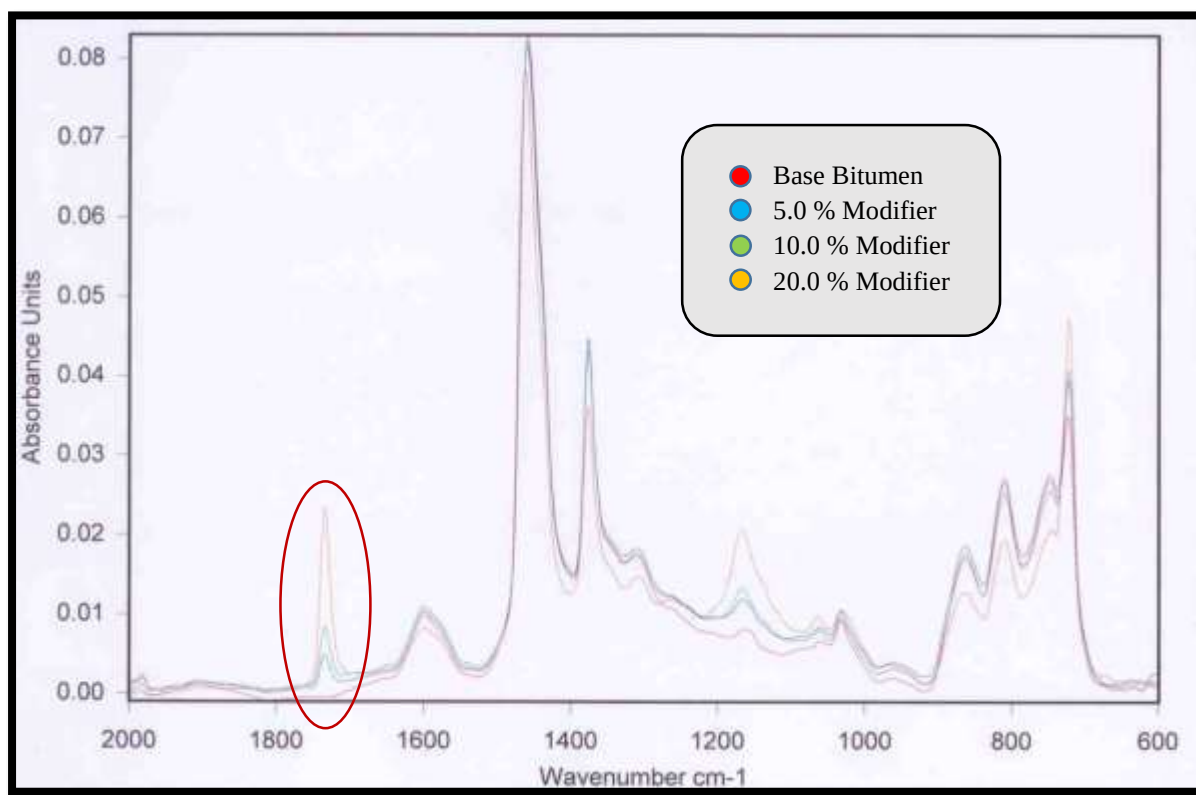
POLYMER INDICATOR FOR ANGLOMAKS 2144



For the Base Bitumen Modified with Anglomaks 2144, the results show that the modifier affects the base bitumen at wavenumber 1734 cm^{-1} . As the amount of bitumen increases, the Absorbance Units also increases. The table shows the relationship between the Absorbance Units and the amount of Modifier.

Amount of Modifier, %	Absorbance Unit(AU)
5.0	0.005
10.0	0.012
20.0	0.028

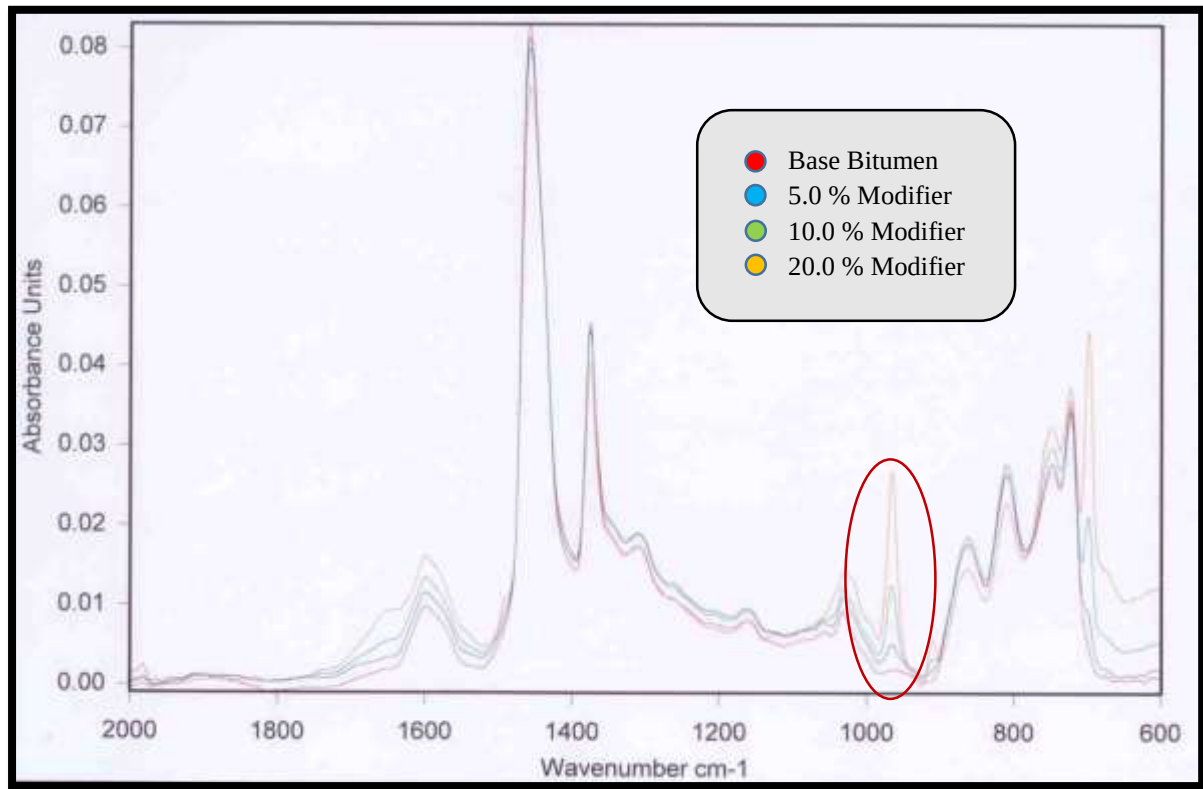
POLYMER INDICATOR FOR LUCOLAST 7010



For the Base Bitumen Modified with Lucolast 7010, the results show that the modifier affects the base bitumen at wavenumber 1734 cm^{-1} which is the same as Anglomak 2144. As the amount of bitumen increases, the Absorbance Units also increases. The table shows the relationship between the Absorbance Units and the amount of Modifier.

Amount of Modifier, %	Absorbance Unit(AU)
5.0	0.005
10.0	0.009
20.0	0.023

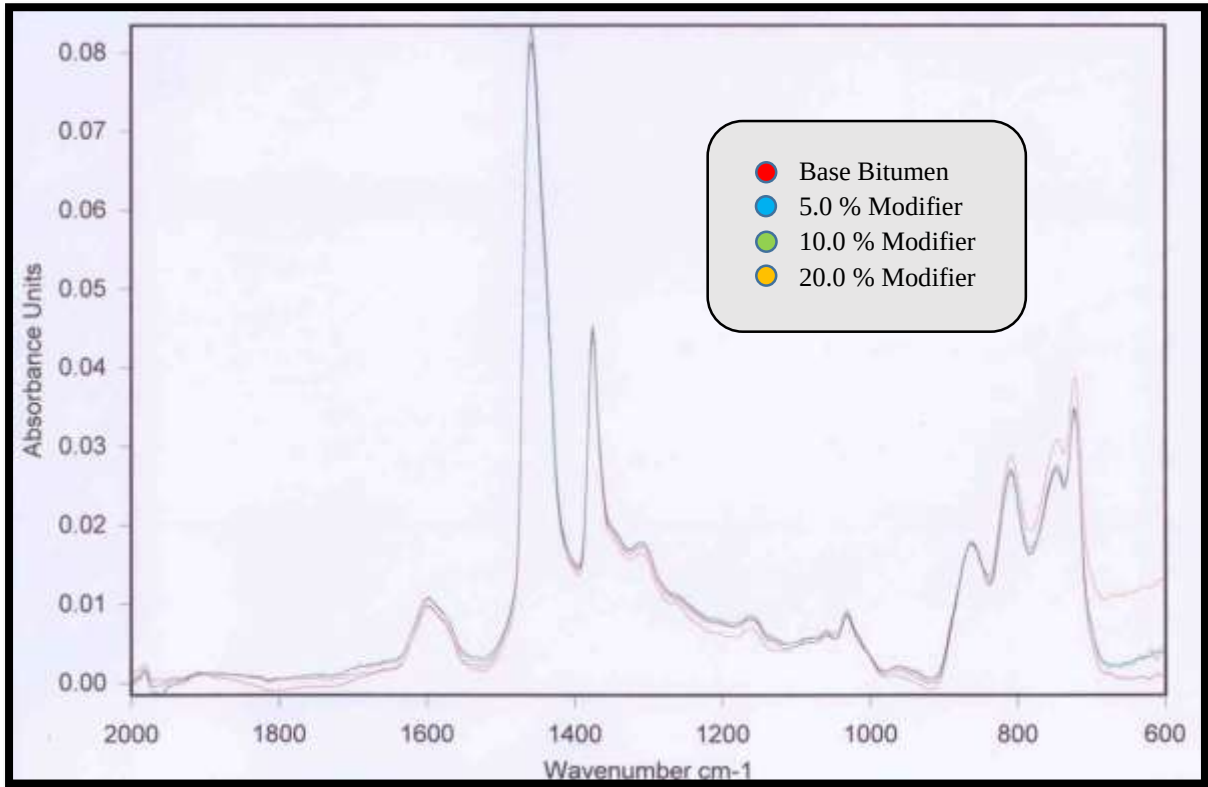
POLYMER INDICATOR FOR SBS KTR 101



For the Base Bitumen Modified with SBS KTR 101, the results show that the modifier affects the base bitumen at wavenumber 966 cm^{-1} . As the amount of bitumen increases, the Absorbance Units also increases. The table shows the relationship between the Absorbance Units and the amount of Modifier.

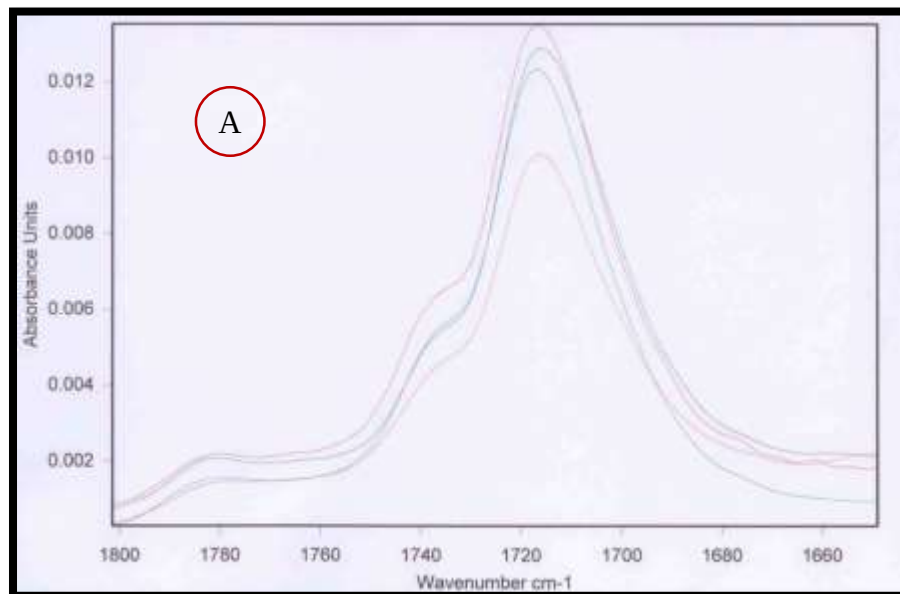
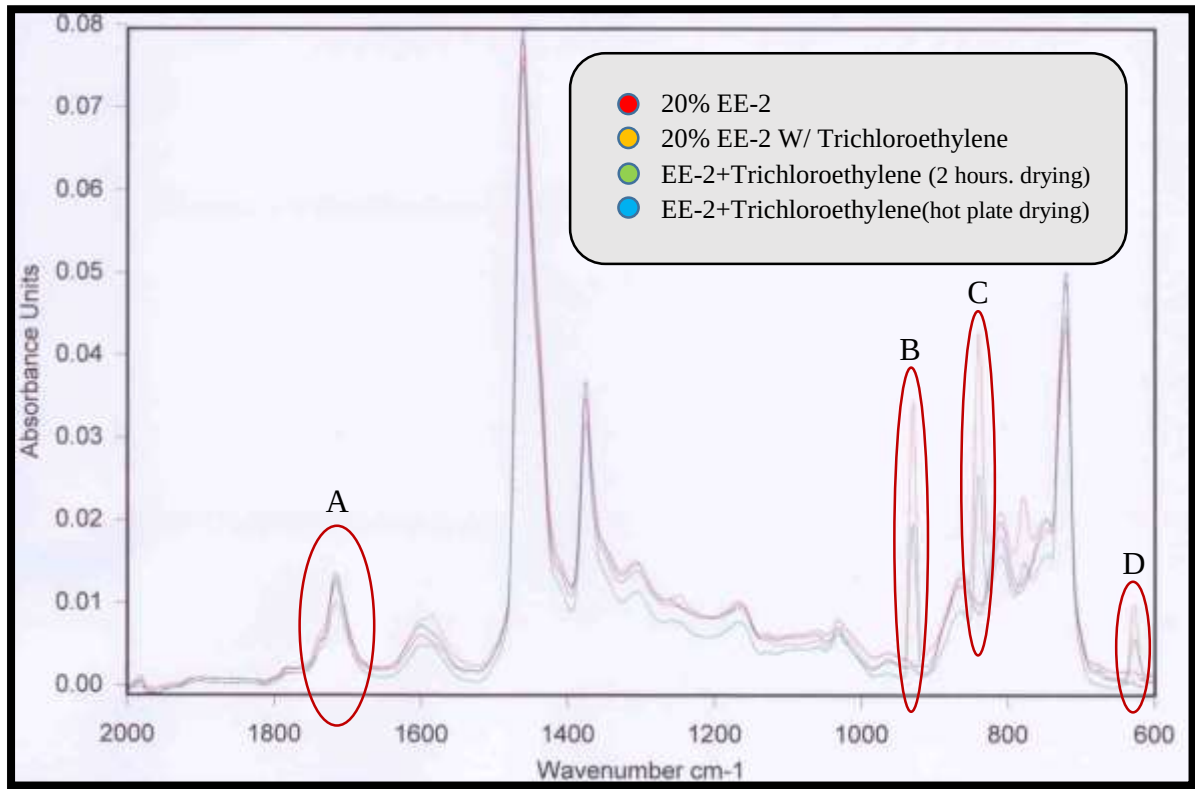
Amount of Modifier, %	Absorbance Unit(AU)
5.0	0.005
10.0	0.013
20.0	0.027

POLYMER INDICATOR FOR CRUMB RUBBER



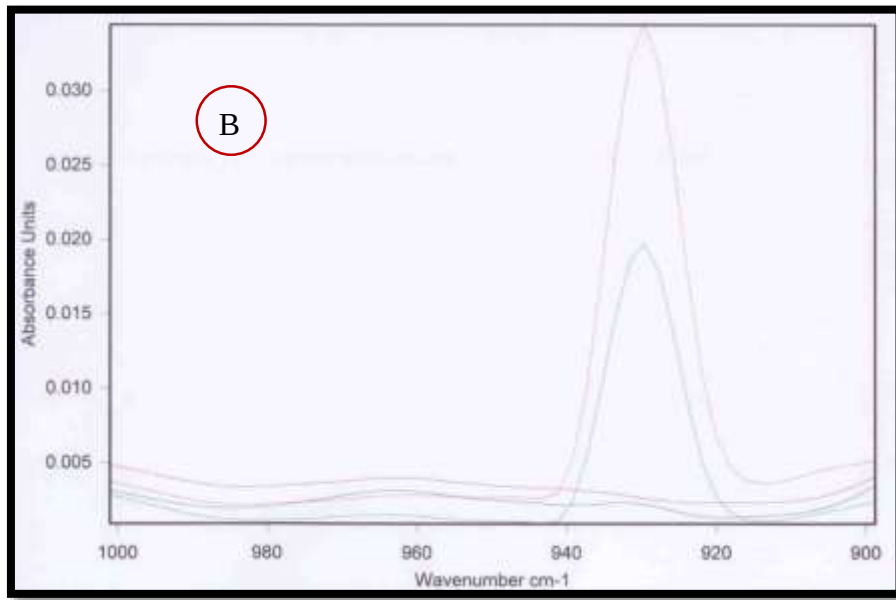
For the Base Bitumen Modified with Crumb Rubber, only passing # 50 particle size of the crumb rubber were used in consideration for the extraction process where a filter paper is used. The results show that the modifier does not affect absorbance spectrum of the base bitumen. The WN/AU spectrum of the crumb rubber at different amount is similar to the WN/AU spectrum of the base bitumen.

EFFECTS OF TRICHLOROETHYLENE ON BITUMEN

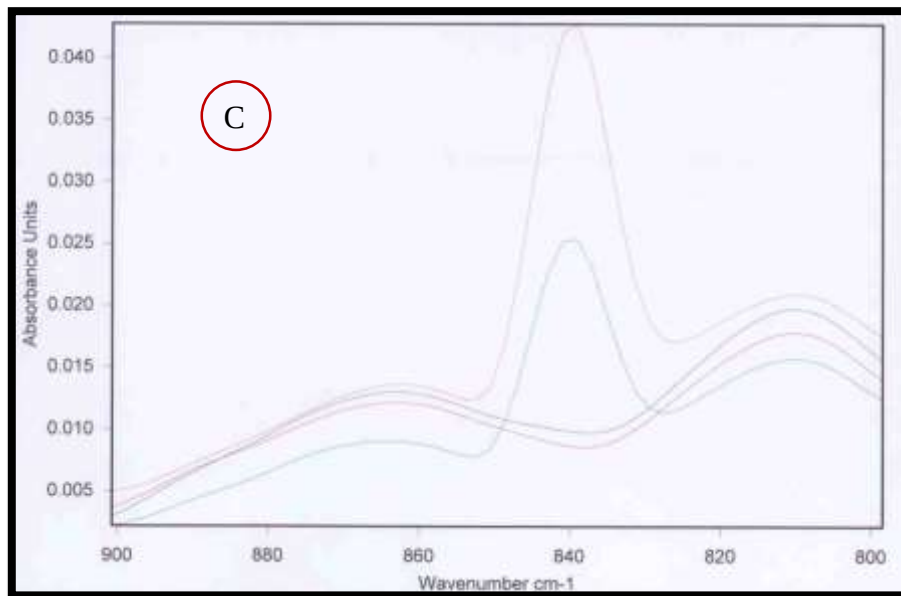


Wavenumber 1800 cm^{-1} - 1650 cm^{-1} shows a decreasing absorbance unit when Trichloroethylene was added to the 20% EE-2 modified bitumen.

- 20% EE-2
- 20% EE-2 W/ Trichloroethylene
- EE-2+Trichloroethylene (2 hours. drying)
- EE-2+Trichloroethylene(hot plate drying)

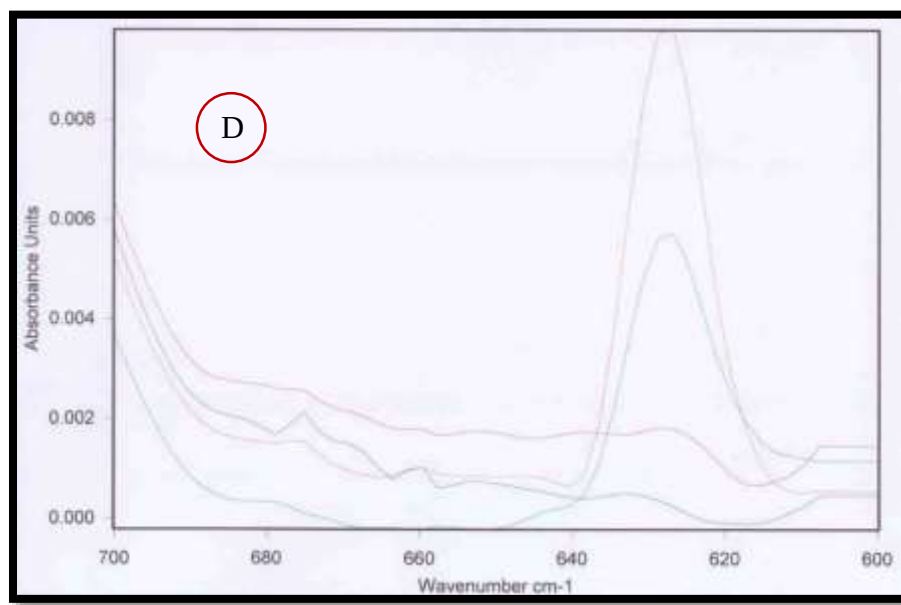


Wavenumber 1000 cm^{-1} - 900 cm^{-1} shows an increasing absorbance unit when Trichloroethylene was added to the 20% EE-2 modified bitumen.



Wavenumber 900 cm^{-1} - 800 cm^{-1} shows an increasing absorbance unit when Trichloroethylene was added to the 20% EE-2 modified bitumen.

- 20% EE-2
- 20% EE-2 W/ Trichloroethylene
- EE-2+Trichloroethylene (2 hours. drying)
- EE-2+Trichloroethylene(hot plate drying)

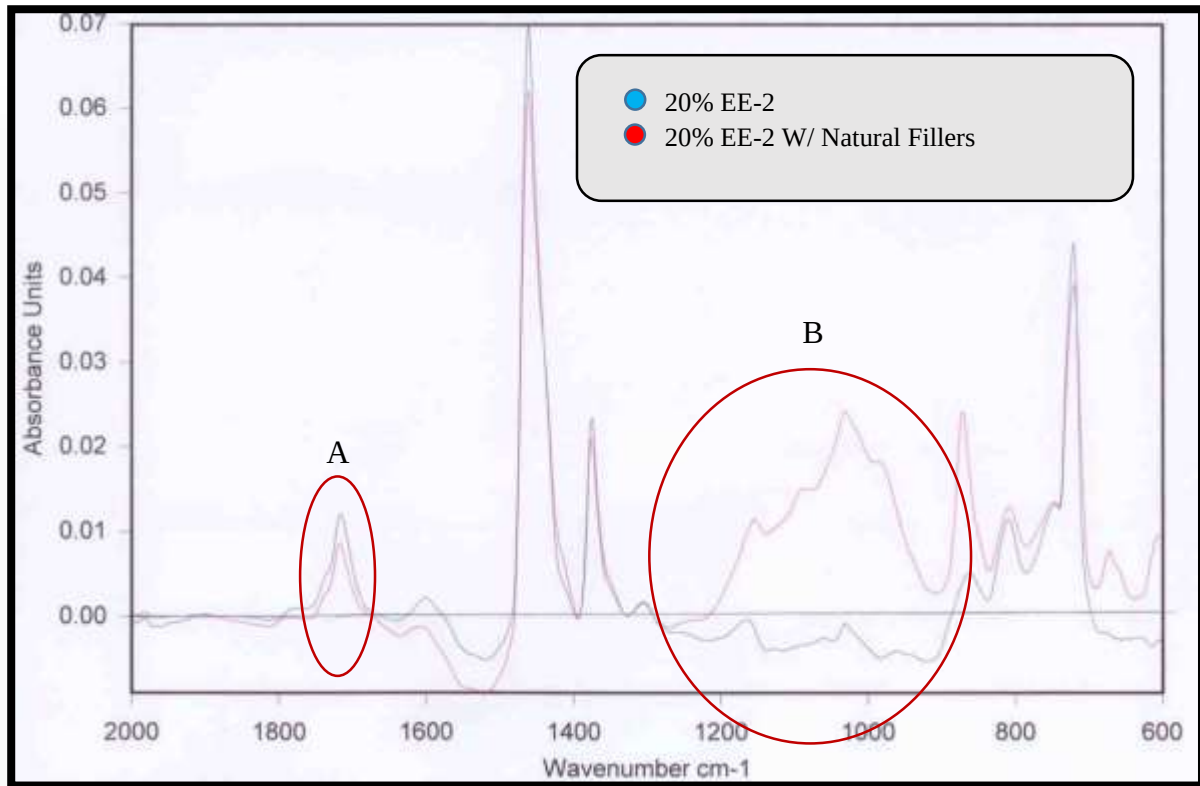


Wavenumber 700 cm^{-1} - 600 cm^{-1} shows an increasing absorbance unit when Trichloroethylene was added to the 20% EE-2 modified bitumen.

A small amount of “20% EE-2 modified bitumen” was prepared and its WN/AU spectrum was established. Small amount of Trichloroethylene was added until the modified bitumen decrease its viscosity then its WN/AU spectrum was also tested. Then the trichloroethylene was removed from the modified bitumen by means of oven drying and hot plate drying.

The WN/AU of the following “20% EE-2 modified bitumen” samples shows that when a Modified bitumen is mixed with Trichloroethylene, there will be an increase of three (3) peak points at wavenumbers 930 cm^{-1} (B), 840 cm^{-1} (C) and 628 cm^{-1} (D). Also, the wavenumber of the EE-2 modified bitumen at 1717 cm^{-1} (A) shows a decrease in absorbance unit when there is a presence of Trichloroethylene in the bitumen sample. This test indicates that the Trichloroethylene in Handing Over samples must be totally removed in order to get an acceptable AU/WN spectrum for comparison with the reference database.

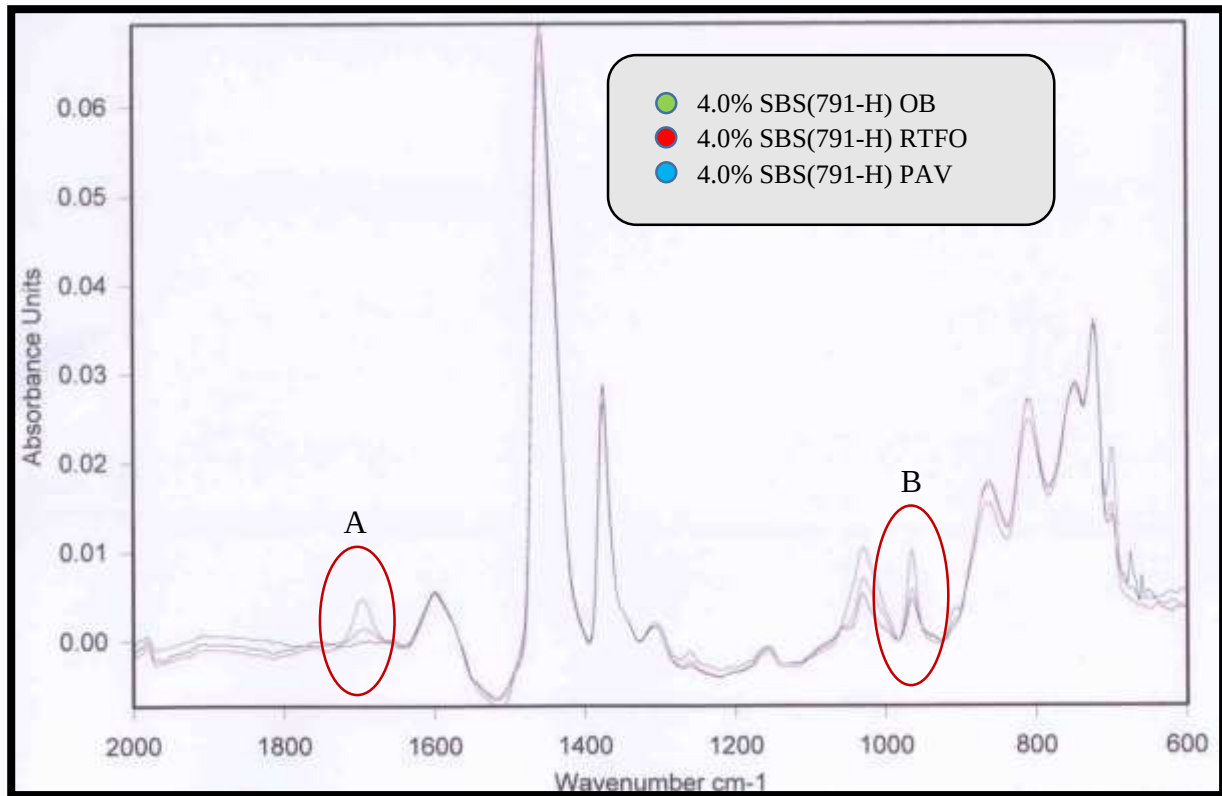
EFFECTS OF NATURAL FILLERS ON BITUMEN



A small amount of “20% EE-2 modified bitumen” was prepared and its WN/AU spectrum was established. Natural Filler was added then its WN/AU spectrum was also tested.

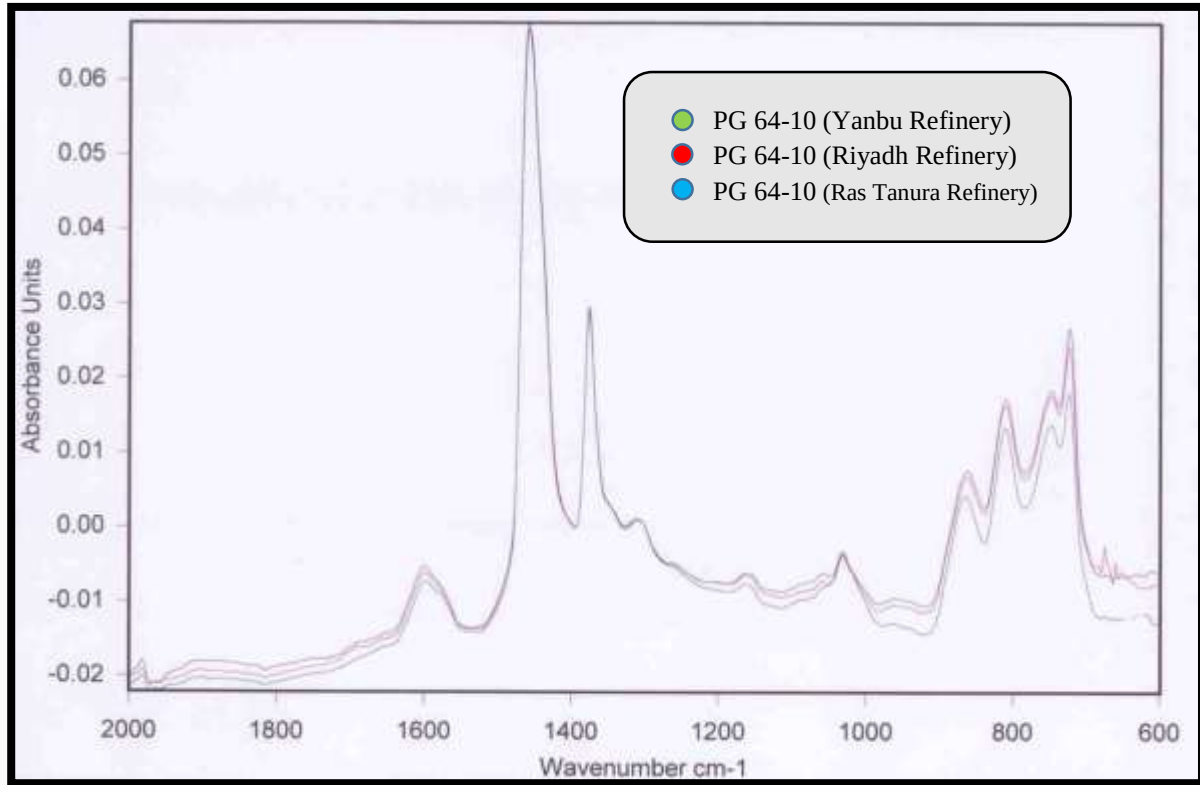
The WN/AU of the following “20% EE-2 modified bitumen” samples shows that when a Modified bitumen is mixed with Natural Filler, there will be an increase of area from wavenumbers 1200 cm^{-1} to 800 cm^{-1} (B). Also, the wavenumber of the EE-2 modified bitumen at 1717 cm^{-1} (A) shows a decrease in absorbance unit when there is a presence of Natural Fillers in the bitumen sample. This test indicates that when a bitumen sample is contaminated with natural fillers, the absorbance unit of the polymer decreases.

AGEING INDICATOR



To determine the wavenumber of the oxidation state or ageing indicator of the bitumen, a 4.0 % SBS (791-H) modified bitumen was prepared and its WN/AU spectrum was taken. Then the sample undergo ageing [RTFO (ASTM D 2872) & PAV (ASTM D 6521)] and its spectrum was also taken for comparison. The results show that ageing or oxidation affects the modified bitumen at wavenumber 1698 cm⁻¹ (A). The peak at wavenumber 966 cm⁻¹ (B) is the SBS polymer indicator. The absorbance unit of the SBS polymer shows an increasing result when the degree of oxidation also increases.

COMPARISON OF BASE BITUMEN FROM DIFFERENT REFINERIES

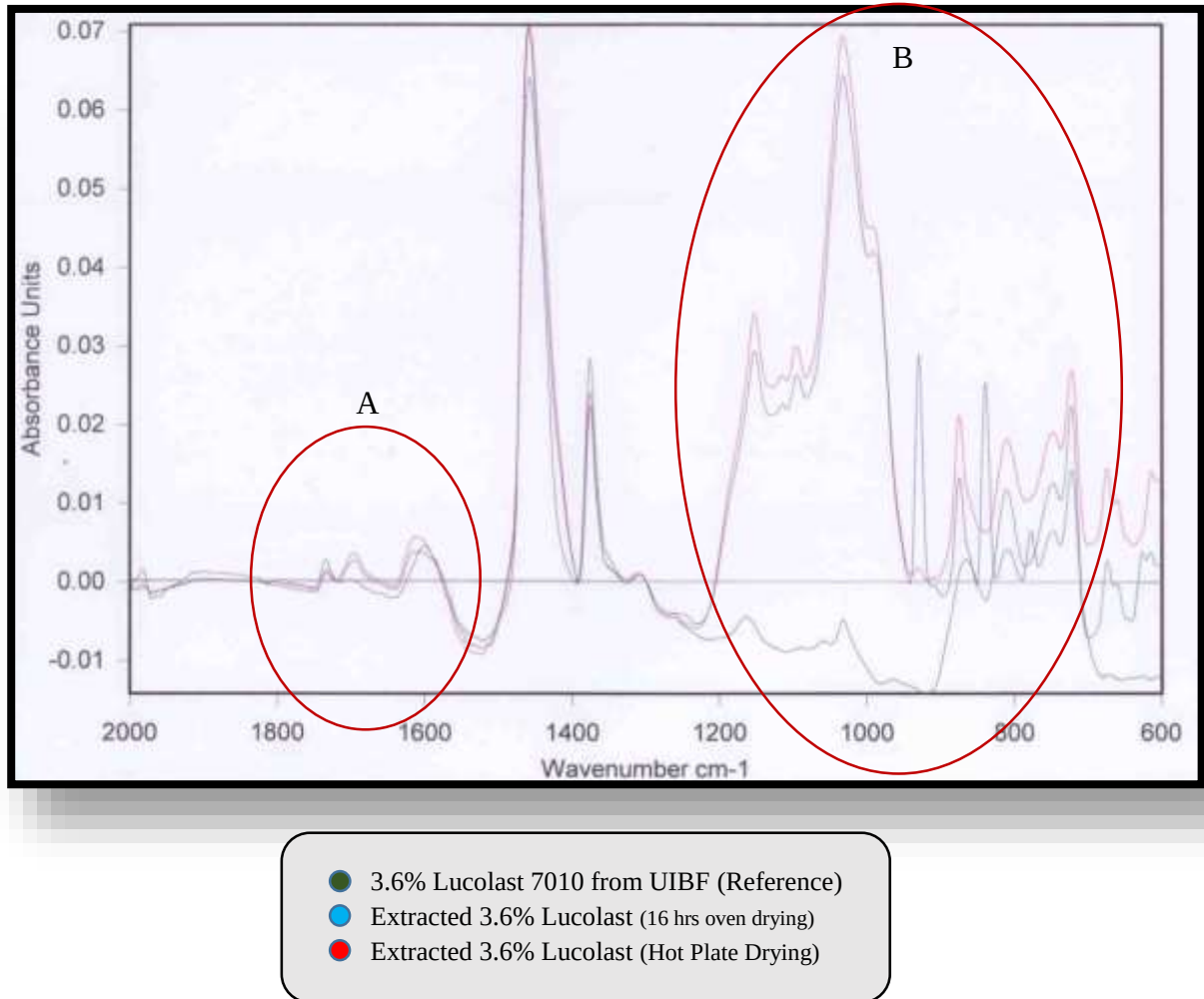


This WN/AU shows the comparison of the different Base Bitumen Refineries. The summary of their correlations is shown at the table below.

	RIYADH	YANBU	RAS TANURA
RIYADH	100 %	98.90 %	99.87 %
YANBU	98.90 %	100 %	98.38 %
RAS TANURA	99.87 %	98.38 %	100 %

These results shows that the bitumen of the different Base Bitumen Refineries can be considered similar to each other.

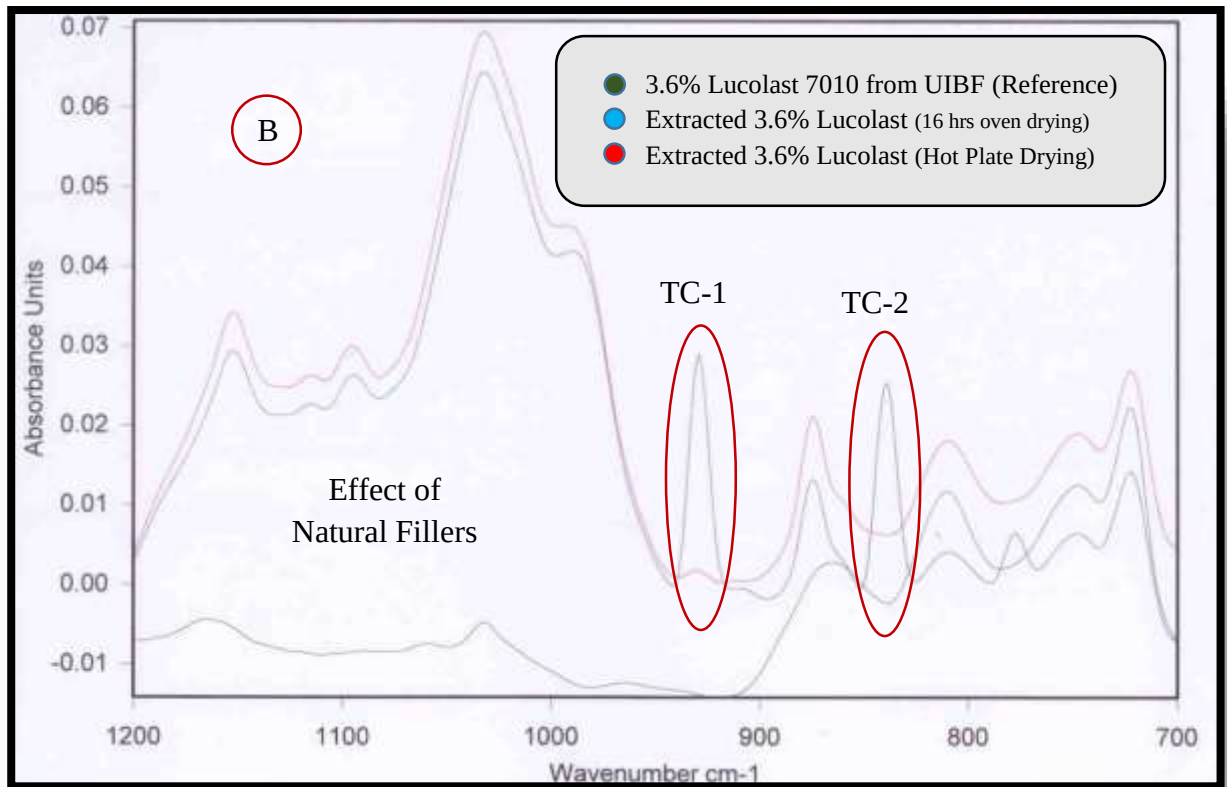
TESTING HANDING OVER SAMPLES



To verify whether the FT-IR is applicable for the Handing Over Sample, the following materials were prepared:

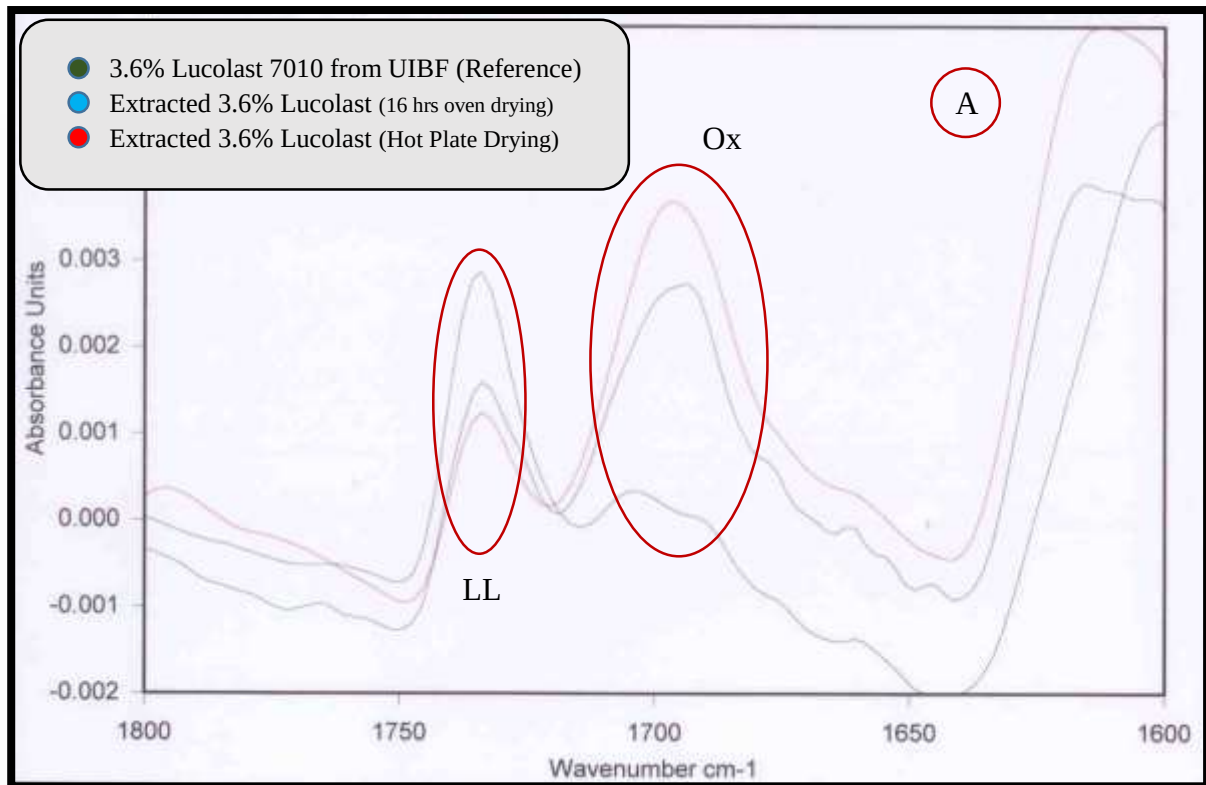
1. Base Bitumen from Riyadh Refinery
2. Base Bitumen from Riyadh Refinery modified with 3.6% Lucolast 7010 (UIBF)
3. 3.6% Lucolast 7010 modified bitumen extracted from loose mix

The first step was to make a WN/AU reference spectrum for Riyadh Base Bitumen and 3.6% Lucolast 7010. Next, the extracted bitumen undergo the process of oven drying to remove the Trichloroethylene.



After 16 hours of drying, the sample was tested to determine if the trichloroethylene was removed. The result shows that there is a huge peak at wavenumbers 930 cm⁻¹ (TC-1) and 840 cm⁻¹ (TC-2) meaning, there is still trichloroethylene present in the sample. Then, the oven dried sample was transferred to a hot plate to continue the drying process of the sample. By observation at wavenumbers 930 cm⁻¹ (TC-1) and 840 cm⁻¹ (TC-2), the size of the peak decreases to a point where there is no more peak. Therefore, the presence of trichloroethylene is no more.

Considering the reference spectrum (3.6% Lucolast 7010 from UIBF), the extracted sample shows an increase in area at wavenumber 1200 cm⁻¹ to 800cm⁻¹. The possible reason for this effect was the presence of natural fillers which are not filtered during the extraction process of the loose mix sample.



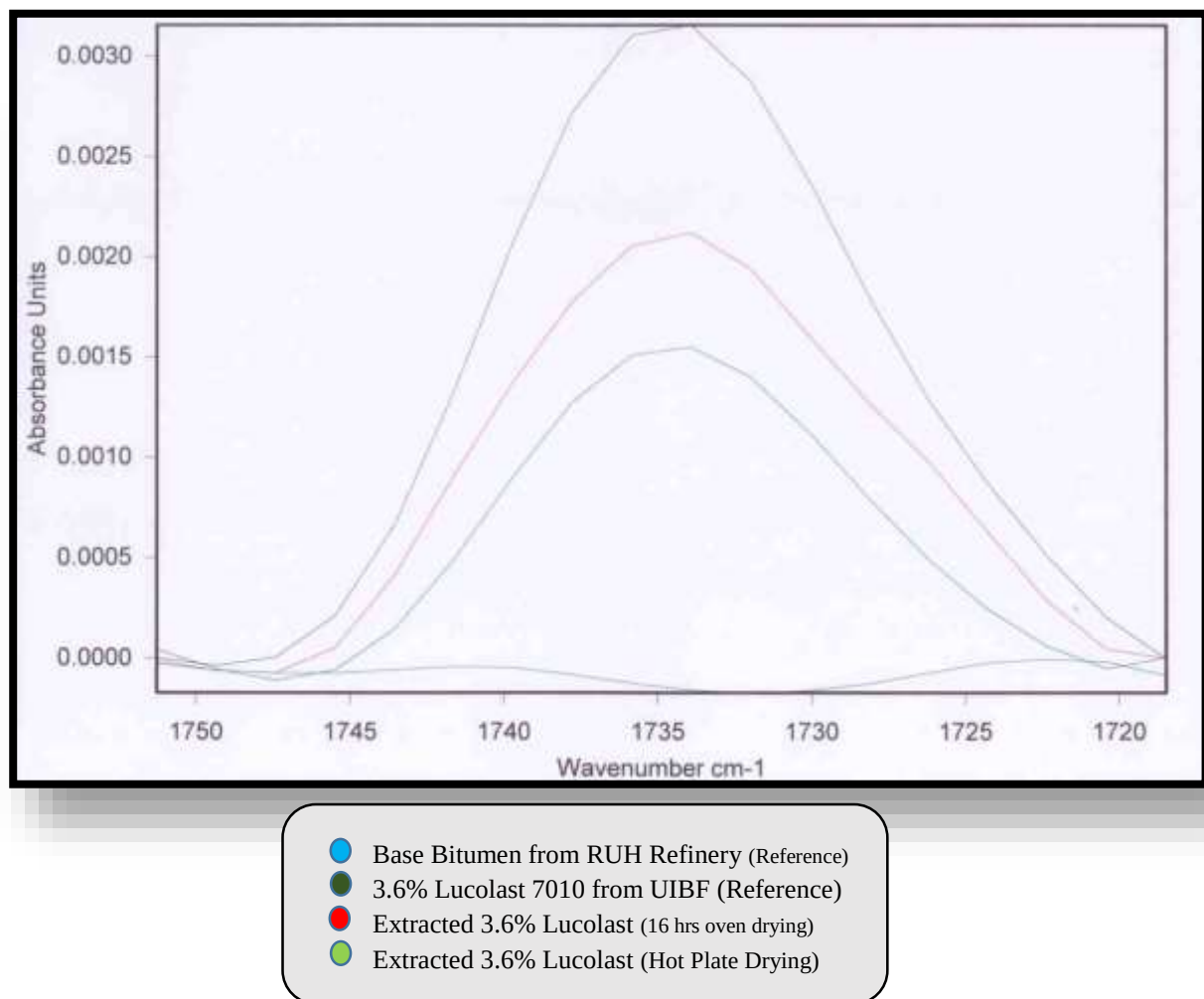
The wavenumber at 1734 cm^{-1} (LL) indicates the presence of Lucolast 7010 in the sample. The wavenumber at 1698 cm^{-1} (Ox) shows the oxidation state/ageing of the sample. With the presence of natural fillers and trichloroethylene, the absorbance unit of the Lucolast decreases in comparison with the reference spectrum. The table below shows the correlation of the extracted sample and the reference spectrum from wavenumber 2000 cm^{-1} to 600 cm^{-1} .

WN/AU (2000 cm^{-1} - 600 cm^{-1})	Reference 3.6% Lucolast 7010	3.6 % Lucolast 7010 Extracted Bitumen (16 hrs oven drying)	3.6 % Lucolast 7010 Extracted Bitumen (hot plate drying)
Reference PG 64-10 (RUH)	98.86 %	51.51 %	53.41 %
Reference 3.6% Lucolast 7010	100 %	50.84 %	54.10 %

The correlation of the Reference Base Bitumen (RUH) to the Reference 3.6 % Lucolast 7010 shows a 98.86 % (passing result – greater than 95 %) which is considered wrong since there is a presence of a polymer mixed with the bitumen.

The results of the correlation using the wavenumber 2000 cm^{-1} - 600 cm^{-1} of the extracted sample in comparison with the reference spectrum shows a failing results (less than 95%). The possible reason for this failure can be the cause of the presence of natural fillers and trichloroethylene which are also considered when correlating the WN/AU spectrum from a large length of the wavenumber (2000 cm^{-1} – 600 cm^{-1}).

Since we are only concern with the presence of the polymer in the extracted bitumen, the length of the wavenumber should only focus on the part where the polymer affects the WN/AU of the spectrum. For this extracted sample, the part where the polymer is active is at wavenumber 1750 cm^{-1} to 1720 cm^{-1} .



WN/AU (1750 cm ⁻¹ -1720 cm ⁻¹)	Reference 3.6% Lucolast 7010	3.6 % Lucolast 7010 Extracted Bitumen (16 hrs oven drying)	3.6 % Lucolast 7010 Extracted Bitumen (hot plate drying)
Reference PG 64-10 (RUH)	0.00 %	0.00 %	0.00 %
Reference 3.6% Lucolast 7010	100 %	99.81 %	99.31 %

The correlation of the Reference Base Bitumen (RUH) to the Reference 3.6 % Lucolast 7010 using smaller length of wavenumber (1750 cm⁻¹ to 1720 cm⁻¹) shows a 0.00 % meaning there are no correlation between this two samples.

The table above shows the results of the correlation using the wavenumber 1750 cm⁻¹ - 1720 cm⁻¹ of the extracted sample in comparison with the reference spectrum which can be considered as an acceptable results.

Chapter 3

RESULTS AND DISCUSSION**Summary of Results:**

Indicators	Wavenumber, cm ⁻¹
EE-2	1717
Anglomaks 2144	1734
Lucolast 7010	1734
SBS KTR 101	966
SBS 791-H	966
Crumb Rubber	Cannot be determined
Trichloroethylene (used for extraction)	930
	840
	628
Natural Filler	1200-800
Ageing / Oxidation State	1698

This Table shows location of the wavenumber where the polymer indicator, ageing indicator and other indicator which affect the bitumen sample.

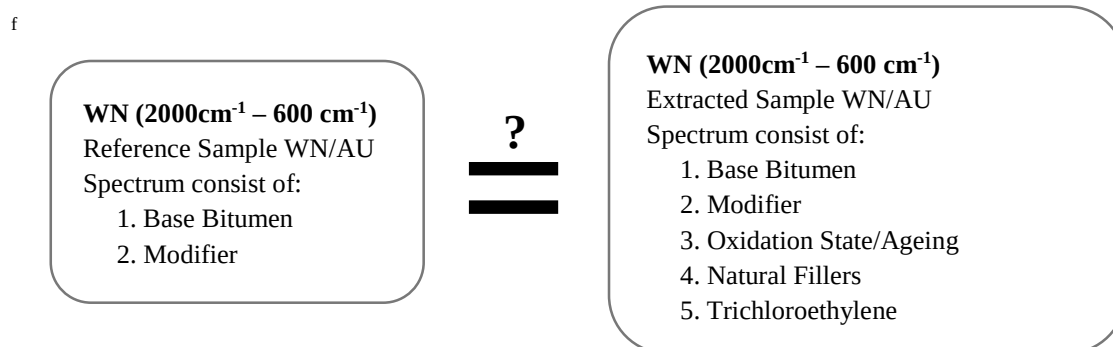
The presence of trichloroethylene and natural filler which are present in the extracted Handing Over samples has an effect on the WN/AU spectrum of the polymer. It is possible for the FT-IR to determine what “kind” of polymer modifier in the bitumen correctly, however, the “amount” of the polymer modifier cannot be determine accurately due to the limitation of the FT-IR PIKE MIRacle™ single-bounce attenuated total reflectance (ATR) cell equipped with a ZnSe single crystal.

Disadvantages of FT-IR is a PIKE MIRacle™ single-bounce attenuated total reflectance (ATR) cell equipped with a ZnSe single crystal

1. According to the Bruker Optics ^[4], highly absorbing substance like carbon black filled rubbers (Crumb Rubbers) are difficult to measure with an ATR crystal because of its low refractive index. To overcome this limitation, Germanium (Ge) crystal is recommended.
2. The ATR method produces poor absorbance ratio relationship to polymer.
3. It is not possible to create a polymer calibration curve to determine the amount of polymer in a bitumen because of its high noise level which influences the absorbance unit of the polymer.

Correlating WN/AU Spectrum

The comparison between the sample and the reference sample spectrum was performed by calculating the correlation coefficient.



When comparing large length of wavenumber (2000 cm⁻¹ to 600 cm⁻¹), we are not just comparing the amount of modifier between the reference spectrum and the sample but the effects of other factors such as oxidation state/ageing, trichloroethylene and natural fillers which were present in the extracted sample are also considered. Therefore, smaller length of wavenumber should be considered since the FT-IR has the capability to focus on the part of the spectrum where the modifier is active.

Chapter 4

CONCLUSION AND RECOMMENDATIONS

1. The presence of the commonly used polymer types in PMABs can be determined by the Bruker Tensor 27 FT-IR Using PIKE MIRacle™ ATR cell. However, this is not applicable for high absorbing substances like Crumb Rubber.
2. The Bruker Tensor 27 FT-IR Using PIKE MIRacle™ ATR cell can distinguish between the commonly used PMABs since the characteristics of the modifier differ from one another.
3. The Bruker Tensor 27 FT-IR Using PIKE MIRacle™ ATR cell cannot effectively determine the amount of polymer present in the PMABS due to the noise level that this apparatus provides.
3. The Bruker Tensor 27 FT-IR Using PIKE MIRacle™ ATR cell can be used as a quality assurance test to detect the presence of polymer type and content in PMABs obtained from the plant tanks and from the Hot-Asphalt mix. However, there are other accessories for the FT-IR which are more effective like the **Germanium (Ge) Crystal** for the ATR cell or **Transmission through a KBr Pellet**.

REFERENCES

1. Bruker Tensor 27 FT-IR Instructions (V1.1)
<http://chem-faculty.lsu.edu/stanley/webpub/Bruker-Tensor-FTIR-instructions.pdf>
2. Bitumen Analysis by FT-IR spectrometry: Testing and Analysis Protocol
by Belgian Road Research Centre
<http://www.brrc.be/publications/me/me8513.pdf>
3. OPUS 6.5 Basics – Manual
http://files.nocnt.ru/hardware/science/senterra/opus65-doc-en/supplement_6.5.pdf
4. Fast and Easy Analysis of Polymeric Samples
http://www.bruker.com/fileadmin/user_upload/8-PDF-Docs/OpticalSpectroscopy/FT-IR/ALPHA/AN/AN84_easy_analysis_of_polymeric_EN.pdf
5. Practical Application of FT-IR to Characterize Paving Materials,
Technical Report 0-5608-1, University of North Texas, December 2010
Nasrazadani, Mielke, Springfield and Naresh